13. $Header\ Linked\ List - Time\ Complexity\ and\ Space\ Complexity - Part\ 2$

1. $Delete\ from\ Beginning$:

- $Time\ Complexity$:

  $\rightarrow Best = Worst = Average = O(1)$.

- $Space\ Complexity$:

  $\rightarrow No\ Parametric\ Input \quad hence, Parametric\ Input\ Space$:

  $Input\ Space = O(1)$.
  $Auxiliary\ Space = O(1)$.

$Total\ Space\ Complexity$: $Input\ Space + Auxiliary = O(1) + O(1) = O(1)$.

$\rightarrow Full\ Data\ Definition$:

$Input\ Space[Considering\ the\ whole\ list] = O(n)$.

$[Deleting\ one\ node\ , total\ nodes\ reduces\ to\ n-1 = O(n-1) = O(n).]$

$Auxiliary\ Space = O(1)$.

$[This\ is\ the\ extra\ space\ or\ temporary\ memory\ used\ by\ the\ algorithm\ to\ complete\ its\ task, excluding\ the\ space\ taken\ by\ the\ input\ data.]$

$Total\ Space\ Complexity$: $Input\ Space + Auxiliary = O(n) + O(1) = O(1)$.

2. *Delete At End*:

- ***Time Complexity***:

  $\rightarrow \boldsymbol{Best} = \Omega(1). \left[\boldsymbol{When\ List\ is\ Empty}\right]$

  $\rightarrow\ \boldsymbol{Worst\ Case} = \boldsymbol{O}(n)\ = \boldsymbol{Average\ Case} = \ \Theta(n).$
  $\left[\begin{array}{c}\boldsymbol{When\ List\ is\ not\ empty\ and\ there\ is\ numerous} \\ \boldsymbol{Nodes\ in\ the\ list.\ Note: True\ Average\ Case\ doesn't} \\ \boldsymbol{exist\ here.}\end{array}\right]$

- ***Space Complexity***:

  → Parametric Input Space:

  There is no parameter present hence:

  $\boldsymbol{Input\ Space} = \boldsymbol{O}(1)\ .$
  $\boldsymbol{Auxiliary\ Space} = \boldsymbol{O}(1).$

$\boldsymbol{Total\ Space\ Complexity}: \boldsymbol{Input\ Space} + \boldsymbol{Auxiliary} = \boldsymbol{O}(1) + \boldsymbol{O}(1) = \boldsymbol{O}(1).$

→ Full Data Definition:

Here Input Space includes `n` no. of elements at `n` no. of nodes and last node gets deleted i.e. $n-1$, then:

$\boldsymbol{Input\ Space} = n - 1 = \boldsymbol{O}(n-1) = \boldsymbol{O}(n)\ .$
$\boldsymbol{Auxiliary\ Space} = \boldsymbol{O}(1).$

$\boldsymbol{Total\ Space\ Complexity}: \boldsymbol{Input\ Space} + \boldsymbol{Auxiliary} = \boldsymbol{O}(n) + \boldsymbol{O}(1) = \boldsymbol{O}(n).$

3. *Delete From Position*:

- ***Time Complexity***:

$$\rightarrow \boldsymbol{Best} = \boldsymbol{\Omega(1)}. \begin{bmatrix} \boldsymbol{When\ List\ is\ Empty} \\ \boldsymbol{Or} \\ \boldsymbol{Pos < 1\ or\ Pos > Head \rightarrow Data} \end{bmatrix}$$

$$\rightarrow\ \boldsymbol{Worst\ Case} = \boldsymbol{O(n)}[\boldsymbol{When\ `pos`\ is\ not\ found.}]$$

→ ***Average Case*** :

→ $If\ the\ position\ is\ 1, the\ loop's\ inner\ statement\ runs\ 0\ time.$

→ $If\ the\ position\ is\ 2, the\ loop's\ inner\ statement\ runs\ 1\ time.$

→ $If\ the\ position\ is\ 3, the\ loop's\ inner\ statement\ runs\ \ 2\ time.$

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→ $If\ the\ position\ is\ (N), the\ loop's\ inner\ statement\ runs\ (N-1)\ times.$

$As\ when\ i = N < N\ \ becomes\ false\ , hence\ no\ execution\ of\ loop's\ inner\ statements.$

$$Hence\ , \sum_{i=1}^{N-1} i = \frac{N(N-1)}{2}.$$

$Now, as\ we\ take\ all\ the\ possibilites\ then\ , we\ take\ the\ best\ case\ also\ i.e.\ O(1), when\ position = \ 1\ . Considering\ `T(P)`\ is\ time\ complexity\ of\ position\ `P`\ .$

$$T(P) = \begin{cases} 1 & , when\ P = 1. \\ P - 1 & , when\ P = 2, 3, 4, .., N. \end{cases}$$

$Now, If\ there\ is\ `N`\ no. of\ nodes\ there\ we\ can\ delete\ data\ at\ \ N\ positions.$

$For\ position\ 1, the\ probability\ of\ deletion\ is \frac{1}{N}.$

$For\ position\ 2, the\ probability\ of\ deletion\ \ is \frac{1}{N}.$

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$For\ position\ N, the\ probability\ of\ deletion\ \ is \frac{1}{N}.$

$Hence\ probability\ of\ deletion\ will\ always\ remain\ \frac{1}{N}.$

$\therefore\ Average\ Time\ Complexity\ \ is:\ probability\ of\ deletion \times (number\ of\ possible\ iterations + constant\ time\ complexities\ for\ left\ over\ positions)$.

$$= \left(\frac{1}{N} \times 1 + \frac{1}{N} \times 2 + \cdots + \frac{1}{N} \times N - 1\right) + \frac{1}{N} \times 1$$

$$= \frac{1}{N} \times \{(1 + 2 + 3 + \cdots + N - 1) + 1\}$$

$$= \frac{1}{N} \times \left\{ \sum_{i=1}^{N-1} i + 1 \right\}$$

$$= \frac{1}{N} \times \left(\frac{N(N-1)}{2} + 1 \right)$$

$$= \frac{N-1}{2} + \frac{1}{N}$$

Appplying Big Theta in the equation:

$$= \Theta\left(\frac{N-1}{2}\right) + \Theta\left(\frac{1}{N}\right)$$

Now, taking out the dominant term of numerator and denominator:

$$= \Theta\left(\frac{N}{2}\right) + \Theta\left(\frac{1}{N}\right)$$

$$= \frac{1}{2} \times \Theta(N) + \Theta\left(\frac{1}{N}\right)$$

Now removing the constant `$\frac{1}{2}$` ***we get***:

$$= \Theta(N) + \Theta\left(\frac{1}{N}\right)$$

When combining two Big Theta Θ **terms , largest growth rate is the result**:

$\rightarrow \Theta(N)$ ***grows linearly with*** `$N$` .

$\rightarrow \Theta\left(\frac{1}{N}\right)$ *decreases to a constant , i.e. when N grows larger ,* $\frac{1}{N}$ *approaches to* 0.

This we can prove by limit and continuity also:

$\lim_{n\to\infty}\left(\frac{1}{n}\right) \Rightarrow \frac{1}{\infty} = 0 \left[\frac{c}{\infty} = 0 \text{ , where c is constant}\right]$.

As $\Theta(N)$*'s growth rate is higher , therefore , average case is* $\Theta(N)$.

Hence, Average Time Complexity is : $\Theta(N)$.

Average Case Time Complexity[*Most Simplest Alternative Apporach*]:

(*Simplest approach*)

For position 1*, the probability of deletion is* $\frac{1}{N}$.

For position 2*, the probability of deletion is* $\frac{1}{N}$.

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For position N*, the probability of deletion is* $\frac{1}{N}$.

Hence probability of deletion will always remain $\frac{1}{N}$*, for position* 1 *to* N.

Average case complexity : $\text{Probability of deletion} \times \begin{pmatrix} \text{Summation} \\ \text{of number of} \\ \text{Position.} \end{pmatrix}$

$$= \frac{1}{N} \times \left(\sum_{i=1}^{N} i \right)$$

$$= \frac{1}{N} \times \left(\frac{N(N+1)}{2} \right)$$

$$= \frac{N+1}{2}$$

$$or, \frac{N}{2} + \frac{1}{2}$$

Appplying Big Theta in the equation:

$$= \Theta\left(\frac{N}{2}\right) + \Theta\left(\frac{1}{2}\right)$$

Now removing the constant `$\frac{1}{2}$` ***we get***:

$$= \Theta(N) + \Theta(1)$$

$$= \Theta(N).$$

$$\textit{Hence, Average Time Complexity is}: \ \Theta\left(\frac{N+1}{2}\right) \textit{ or } \Theta(N).$$

- ***Space Complexity***:

  → Parametric Input Space:

  There is a parameter `int pos` present hence:

  $$\textbf{\textit{Input Space}} = \boldsymbol{O(1)}\ .$$
  $$\textbf{\textit{Auxiliary Space}} = \boldsymbol{O(1)}.$$

$\textbf{\textit{Total Space Complexity}}: \textbf{\textit{Input Space}} + \textbf{\textit{Auxiliary}} = \boldsymbol{O(1) + O(1) = O(1)}.$

→ Full Data Definition:

Here Input Space includes `n` no. of elements at `n` no. of nodes and if a node gets deleted i.e. $\boldsymbol{n-1}$, ***then***:

$$\textbf{\textit{Input Space}} = \boldsymbol{n - 1 = O(n-1) = O(n)}\ .$$
$$\textbf{\textit{Auxiliary Space}} = \boldsymbol{O(1)}.$$

$\textbf{\textit{Total Space Complexity}}: \textbf{\textit{Input Space}} + \textbf{\textit{Auxiliary}} = \boldsymbol{O(n) + O(1) = O(n)}.$

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4. *Destroy List*:

- ***Time Complexity***:

  $\rightarrow \boldsymbol{Best} = \boldsymbol{\Omega(1)}. \left[\boldsymbol{When\ List\ is\ Empty}\right]$

  $\rightarrow \boldsymbol{Worst\ Case} = \boldsymbol{O(n)} = \boldsymbol{Average\ Case} = \boldsymbol{\Theta(n)}.$
  $\left[\begin{array}{c}\boldsymbol{When\ List\ is\ not\ empty\ and\ there\ is\ numerous} \\ \boldsymbol{Nodes\ in\ the\ list.\ Note: True\ Average\ Case\ doesn't} \\ \boldsymbol{exist\ here.}\end{array}\right]$

- ***Space Complexity***:

  → Parametric Input Space:

  There is no parameter present hence:

  $\boldsymbol{Input\ Space} = \boldsymbol{O(1)}$.
  $\boldsymbol{Auxiliary\ Space} = \boldsymbol{O(1)}$.

$\boldsymbol{Total\ Space\ Complexity}: \boldsymbol{Input\ Space} + \boldsymbol{Auxiliary} = \boldsymbol{O(1)} + \boldsymbol{O(1)} = \boldsymbol{O(1)}$.

→ Full Data Definition:

Before Deletion:

$\boldsymbol{List\ `n`\ nodes} \rightarrow \boldsymbol{O(n)}$. $[\boldsymbol{Input\ Space}]$

$\boldsymbol{Auxiliary\ Space} \rightarrow \boldsymbol{O(1)}$.

$\boldsymbol{Total\ Space\ Complexity}: \boldsymbol{Input\ Space} + \boldsymbol{Auxiliary} = \boldsymbol{O(n)} + \boldsymbol{O(1)}$
$= \boldsymbol{O(n)}$.

After Deletion:

- ***List becomes empty*** $\rightarrow \boldsymbol{O(1)}$. [***Input Space***]
- ***Auxiliary Space*** $\rightarrow \boldsymbol{O(1)}$.

Total Space Complexity: ***Input Space*** + ***Auxiliary*** $= \boldsymbol{O(1)} + \boldsymbol{O(1)} = \boldsymbol{O(1)}$.

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5. ***Free Header***:

- ***Time Complexity***:

  $\rightarrow$ ***Best*** $=$ ***Worst*** $=$ ***Average*** $= \boldsymbol{O(1)}$.

- ***Space Complexity***:

$\rightarrow$ ***For both, Parametric Input Space and Full Data Definition***:

Input Space $= \boldsymbol{O(1)}$.
Auxiliary Space $= \boldsymbol{O(1)}$.

[***As,*** ***DestroyList*****() *function already executed.***]

Total Space Complexity: ***Input Space*** + ***Auxiliary*** $= \boldsymbol{O(1)} + \boldsymbol{O(1)} = \boldsymbol{O(1)}$.
